



Homework 2

Regression · Quarto Blogging

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Directions

- Submit your **Quarto document** for **Part 1** to Brightspace using this filename format:
 - `dan1-320-hw2-LASTNAME-FIRSTNAME.qmd`
(e.g., `dan1-320-hw2-choe-byeonghak.qmd`)
- **Due:** March 2, 2026, 3:30 PM (ET)
- Questions? Email Byeong-Hak bchoe@geneseo.edu.

Required R Packages

```
library(tidyverse)
library(broom)
library(stargazer)
library(rmarkdown)
library(margins)
library(yardstick)
library(wvPlots)
library(pROC)
```

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Part 1. Regression



Consider the `homes` DataFrame from the 2004 American Housing Survey, which includes data on home values, demographics, schools, income, finance, mortgages, sales, neighborhood characteristics, noise, smells, state geography, and urban classification.

```
homes <- read_csv(  
  'https://bcdanl.github.io/data/american_housing_survey.csv'  
)  
  
homes <- homes |>  
  mutate(GT20DWN = ifelse((LPRICE - AMMORT) / LPRICE > 0.2,  
                           1, 0)  
)
```

Show

10

 entries

Search:

	LPRICE	VALUE	STATE	METRO	ZINC2	HHGRAD	BATHS	BEDRMS
1	85000	150000	GA	rural	15600	No HS	2	3
2	76500	130000	GA	rural	61001	HS Grad	2	3
3	93900	135000	GA	rural	38700	HS Grad	2	3
4	100000	140000	GA	rural	80000	No HS	3	4
5	100000	135000	GA	ru	000	HS Grad	2	3
6	96000	125000	GA	rural	265000	Bach	2	3

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	LPRICE ▴ ▾	VALUE ▴ ▾	STATE ▴ ▾	METRO ▴ ▾	ZINC2 ▴ ▾	HHGRAD ▴ ▾	BATHS ▴ ▾	BEDRMS ▴ ▾
7	130500	135000	GA	rural	42000	Bach	2	3
8	120000	130000	GA	rural	87000	HS Grad	2	4
9	199900	250000	GA	rural	87800	No HS	3	3
10	105000	110000	GA	rural	27030	No HS	2	3

Showing 1 to 10 of 15,565 entries

Previous

1

2

3

4

5

...

1,557

Next

Variable Description

Variable	Description
LPRICE	Purchase price of unit and land
VALUE	Current market value of unit
STATE	State code
METRO	Central city/suburban status
ZINC2	Household income
HHGRAD	Educational level of householder
BATHS	Number of full bathrooms in unit
BEDRMS	Number of bedrooms in unit
PER	Number of persons in household
ZADULT	Number of adults (18+) in household
NUNITS	Number of units in building
EAPTBL	Apartment buildings within 1/2 block of unit
ECOM1	Business/institutions within 1/2 block
ECOM2	Factories/other industry within 1/2 block
EGREEN	Open spaces within 1/2 block of unit

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Variable	Description
EJUNK	Trash/junk in streets/properties in 1/2 block
ELOW1	Single-family town/rowhouses in 1/2 block
ESFD	Single-family homes within 1/2 block
ETRANS	RR/airport/4-lane highway within 1/2 block
EABAN	Abandoned/vandalized buildings within 1/2 block
HOWH	Rating of unit as a place to live
HOWN	Rating of neighborhood as a place to live
ODORA	Neighborhood has bad smells
STRNA	Neighborhood has heavy street noise/traffic
FRSTHO	First home
AMMORT	Amount of 1st mortgage when acquired
INTW	Interest rate of 1st mortgage (whole number %)
MATBUY	Got 1st mortgage in the same year bought unit
DWNPAY	Main source of down payment on unit

Question 1

Plot some relationships and tell a story.

Show answer

```
# install.packages(c("ggcorrheatmap", "fastDummies"))
library(ggcorrheatmap)
library(fastDummies)

homes_dummy <- homes |>
  select(-STATE) |>
  dummy_cols() |>
  select_if(is.numeric)

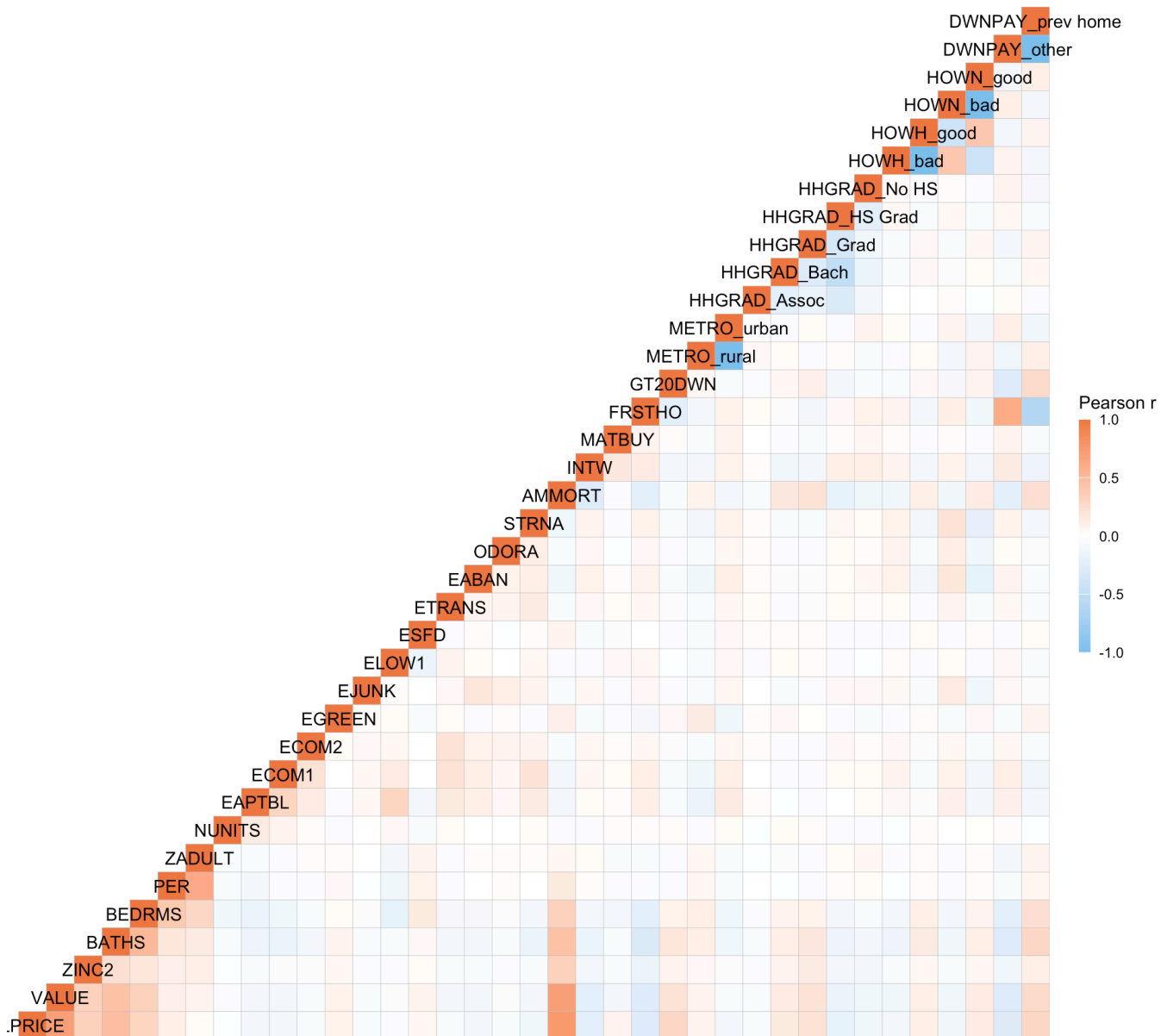
ggcorrhm(homes_dummy,
  layout = "bottomright") +
```

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```

guides(fill = guide_colorbar(
  theme = theme(
    legend.key.width = unit(0.5, "lines"),
    legend.key.height = unit(10, "lines")
  ))
)

```



- `LPRICE / VALUE` and `AMMORT` are positive correlated.
- `FRSTHO` and `DWNPAY_other` are positively correlated.
- `FRSTHO` and `DWNPAY_prev_home` are negatively correlated.

Note

- More details on `ggcorrheatmap` is: <https://lucdr.github.io/ggcorrheatmap/>

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Question 2

- Fit a **linear regression model** with the following specifications:
 - Outcome variable:** $\log(\text{VALUE})$
 - Predictors:** all but **AMMORT** and **LPRICE**

Show answer

Checking-out Data

```
glimpse(homes)
```

Rows: 15,565

Columns: 30

```
$ LPRICE <dbl> 85000, 76500, 93900, 100000, 100000, 96000, 130500, 120000, 19...
$ VALUE <dbl> 150000, 130000, 135000, 140000, 135000, 125000, 135000, 130000...
$ STATE <chr> "GA", "GA", "GA", "GA", "GA", "GA", "GA", "GA", "GA", "GA", "G...
$ METRO <chr> "rural", "rural", "rural", "rural", "rural", "rural", "rural", "rural",...
$ ZINC2 <dbl> 15600, 61001, 38700, 80000, 61000, 265000, 42000, 87000, 87800...
$ HHGRAD <chr> "No HS", "HS Grad", "HS Grad", "No HS", "HS Grad", "Bach", "Ba...
$ BATHS <dbl> 2, 2, 2, 3, 2, 2, 2, 2, 3, 2, 2, 1, 2, 2, 2, 2, 2, 2, 1, 3,...
$ BEDRMS <dbl> 3, 3, 3, 4, 3, 3, 3, 4, 3, 3, 3, 3, 4, 3, 3, 3, 3, 3, 4, 5,...
$ PER <dbl> 1, 5, 4, 2, 2, 2, 1, 2, 4, 3, 2, 4, 4, 3, 4, 1, 3, 1, 1, 5, 5,...
$ ZADULT <dbl> 1, 2, 2, 2, 2, 2, 1, 2, 2, 1, 2, 2, 3, 3, 3, 1, 2, 1, 1, 4, 2,...
$ NUNITS <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
$ EAPTBL <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ ECOM1 <dbl> 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,...
$ ECOM2 <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ EGREEN <dbl> 1, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0,...
$ EJUNK <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ ELOW1 <dbl> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ ESFD <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1,...
$ ETRANS <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,...
$ EABAN <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ HOWH <chr> "good", "good", "good", "good", "good", "good", "good", "good", "good"...
$ HOWN <chr> "good", "bad", "good", "good", "good", "good", "good", "good", "good",...
$ ODORA <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ STRNA <dbl> 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ AMMORT <dbl> 50000, 70000, 117000, 100000, 100000, 96000, 130500, 120000, 1...
$ INTW <dbl> 9, 5, 6, 7, 4, 7, 5, 5, 5, 4, 6, 6, 6, 5, 9, 5, 5, 5, 5, 5, 6,...
$ MATBUY <dbl> 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 0, 0,...
$ DWNPAY <chr> "other", "other", "other", "prev home", "other", "prev home", ...
$ FRSTHO <dbl> 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0,...
$ GT20DWN <dbl> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1,...
```

```
homes |>
count(STATE)
```

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```
# A tibble: 13 × 2
  STATE      n
  <chr> <int>
1 CA      1426
2 CO      1807
3 CT      1422
4 GA      1457
5 IL       237
6 IN      1492
7 LA       779
8 MO      1047
9 OH      1189
10 OK      1173
11 PA      1039
12 TX      1094
13 WA      1403
```

```
homes |>
  count(METRO)
```

```
# A tibble: 2 × 2
  METRO      n
  <chr> <int>
1 rural 12140
2 urban  3425
```

```
homes |>
  count(BEDRMS) # can be either continuous or categorical.
```

```
# A tibble: 9 × 2
  BEDRMS      n
  <dbl> <int>
1     0     2
2     1    197
3     2   2210
4     3   7888
5     4   4322
6     5    812
7     6    114
8     7     16
9     8      4
```

```
homes |>
  count(HOWH)
```

```
# A tibble: 2 × 2
  HOWH      n
  <chr> <int>
1 bad   1275
2 good 14290
```

```
homes |>
  count(HOWN)
```

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```
# A tibble: 2 × 2
  HOWN      n
<chr> <int>
1 bad    2065
2 good  13500
```

```
homes |>
  count(DWNPAY)
```

```
# A tibble: 2 × 2
  DWNPAY      n
<chr>    <int>
1 other   10272
2 prev home 5293
```

Training-Test Split

```
set.seed(1)
homes <- homes |>
  mutate(rnd = runif(n())) |>
  mutate(
    BEDRMS_cat = case_when(
      BEDRMS <= 1 ~ "1 or fewer",
      BEDRMS == 2 ~ "2",
      BEDRMS == 3 ~ "3",
      BEDRMS == 4 ~ "4",
      BEDRMS >= 5 ~ "5+"
    ),
    BEDRMS_cat = factor(BEDRMS_cat)
  )

dtrain <- homes |>
  filter(rnd > .25) |>
  select(-rnd)

dtest <- homes |>
  filter(rnd <= .25) |>
  select(-rnd)
```

Fitting the Model

```
m1 <- lm(log(VALUE) ~ .,
  data = dtrain |> select(-AMMORT, -LPRICE, -GT20DWN, -BEDRMS)
)

stargazer(m1, type = "html")
```

Dependent variable:	
	log(VALUE)
STATECO	-0.304*** (0.034)

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STATECT	-0.359*** (0.035)
STATEGA	-0.659*** (0.036)
STATEIL	-0.879*** (0.067)
STATEIN	-0.799*** (0.035)
STATELA	-0.746*** (0.042)
STATEMO	-0.691*** (0.039)
STATEOH	-0.698*** (0.038)
STATEOK	-1.030*** (0.037)
STATEPA	-0.896*** (0.039)
STATETX	-1.084*** (0.039)
STATEWA	-0.126*** (0.035)
METROurban	0.085*** (0.021)
ZINC2	0.00000*** (0.00000)
HHGRADBach	0.129*** (0.026)
HHGRADGrad	0.213*** (0.030)
HHGRADHS Grad	-0.065*** (0.025)
HHGRADNo HS	-0.165*** (0.037)
BATHS	0.205*** (0.013)
PER	0.013* (0.007)
ZADULT	-0.026** (0.012)
NUNITS	-0.001** (0.001)
EAPTBL	-0.035

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	(0.027)
ECOM1	-0.015
	(0.022)
ECOM2	-0.094 [*]
	(0.055)
EGREEN	0.008
	(0.016)
EJUNK	-0.150 ^{**}
	(0.058)
ELOW1	0.046 [*]
	(0.027)
ESFD	0.327 ^{***}
	(0.034)
ETRANS	-0.011
	(0.029)
EABAN	-0.180 ^{***}
	(0.041)
HOWHgood	0.148 ^{***}
	(0.030)
HOWNgood	0.107 ^{***}
	(0.025)
ODORA	0.010
	(0.038)
STRNA	-0.040 ^{**}
	(0.018)
INTW	-0.043 ^{***}
	(0.005)
MATBUY	-0.027 [*]
	(0.016)
DWNPAYprev home	0.133 ^{***}
	(0.020)
FRSTHO	-0.091 ^{***}
	(0.020)
BEDRMS_cat2	0.199 ^{***}
	(0.072)
BEDRMS_cat3	0.297 ^{***}
	(0.071)
BEDRMS_cat4	0.388 ^{***}
	(0.073)
BEDRMS_cat5+	0.450 ^{***}
	(0.080)
Constant	11.542 ^{***}
	(0.096)

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Observations	11,690
R ²	0.311
Adjusted R ²	0.309
Residual Std. Error	0.812 (df = 11646)
F Statistic	122.331*** (df = 43; 11646)
Note: $p < 0.1$; $p < 0.05$; $p < 0.01$	

Question 3

- Refit the **linear regression model**, retaining only statistically significant predictors from Question 2.
- Compare the revised model to the initial model from **Question 2** using:
 - β estimates
 - R^2
 - RMSE
 - Residual plots

Show answer

```
m1_beta <- m1 |>
  tidy(conf.int = T) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "***", # smallest p-values first
      p.value < .05 ~ "**",
      p.value < .1 ~ "*",
      TRUE ~ ""
    ),
    .after = estimate
  )

m1_beta_nostars <- m1_beta |>
  filter(stars == "")

m2 <- lm(log(VALUE) ~ .,
  data = dtrain |> select(-AMMORT, -LPRICE, -GT20DWN, -BEDRMS,
    -m1_beta_nostars$term)
)

stargazer(m1, m2, type = "html")
```

Dependent variable:

	(1)	log(VALUE)
STATECO	-0.304***	-0.303***

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	(0.034)	(0.034)
STATECT	-0.359***	-0.358***
	(0.035)	(0.035)
STATEGA	-0.659***	-0.660***
	(0.036)	(0.035)
STATEIL	-0.879***	-0.880***
	(0.067)	(0.067)
STATEIN	-0.799***	-0.798***
	(0.035)	(0.035)
STATELA	-0.746***	-0.748***
	(0.042)	(0.042)
STATEMO	-0.691***	-0.692***
	(0.039)	(0.038)
STATEOH	-0.698***	-0.700***
	(0.038)	(0.037)
STATEOK	-1.030***	-1.027***
	(0.037)	(0.037)
STATEPA	-0.896***	-0.895***
	(0.039)	(0.039)
STATETX	-1.084***	-1.082***
	(0.039)	(0.039)
STATEWA	-0.126***	-0.126***
	(0.035)	(0.035)
METROurban	0.085***	0.080***
	(0.021)	(0.020)
ZINC2	0.00000***	0.00000***
	(0.00000)	(0.00000)
HHGRADBach	0.129***	0.130***
	(0.026)	(0.026)
HHGRADGrad	0.213***	0.214***
	(0.030)	(0.030)
HHGRADHS Grad	-0.065***	-0.065***
	(0.025)	(0.025)
HHGRADNo HS	-0.165***	-0.166***
	(0.037)	(0.037)
BATHS	0.205***	0.206***
	(0.013)	(0.013)
PER	0.013*	0.013*
	(0.007)	(0.007)
ZADULT	-0.026**	-0.026**
	(0.012)	(0.012)
NUNITS	-0.001**	-0.001**
	(0.001)	(0.001)

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EAPTBL	-0.035 (0.027)	
ECOM1	-0.015 (0.022)	
ECOM2	-0.094* (0.055)	-0.108** (0.053)
EGREEN	0.008 (0.016)	
EJUNK	-0.150** (0.058)	-0.150*** (0.058)
ELOW1	0.046* (0.027)	0.036 (0.026)
ESFD	0.327*** (0.034)	0.327*** (0.034)
ETRANS	-0.011 (0.029)	
EABAN	-0.180*** (0.041)	-0.184*** (0.041)
HOWHgood	0.148*** (0.030)	0.146*** (0.030)
HOWNgood	0.107*** (0.025)	0.110*** (0.025)
ODORA	0.010 (0.038)	
STRNA	-0.040** (0.018)	-0.044** (0.018)
INTW	-0.043*** (0.005)	-0.043*** (0.005)
MATBUY	-0.027* (0.016)	-0.027* (0.016)
DWNPAYprev home	0.133*** (0.020)	0.134*** (0.020)
FRSTHO	-0.091*** (0.020)	-0.093*** (0.020)
BEDRMS_cat2	0.199*** (0.072)	0.206*** (0.071)
BEDRMS_cat3	0.297*** (0.071)	0.307*** (0.071)
BEDRMS_cat4	0.388*** (0.073)	0.398*** (0.073)
BEDRMS_cat5+	0.450*** (0.080)	

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Constant	11.542*** (0.096)	11.531*** (0.095)
Observations	11,690	11,690
R ²	0.311	0.311
Adjusted R ²	0.309	0.309
Residual Std. Error	0.812 (df = 11646)	0.812 (df = 11651)
F Statistic	122.331*** (df = 43; 11646)	138.363*** (df = 38; 11651)
Note:		$p < 0.1$; $p < 0.05$; $p < 0.01$

β estimates

```

m2_beta <- m2 |>
  tidy(conf.int = T) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "****", # smallest p-values first
      p.value < .05 ~ "***",
      p.value < .1 ~ "**",
      TRUE ~ ""
    ),
    .after = estimate
  )

m1_beta <- m1_beta |>
  select(term, estimate, stars, starts_with("conf")) |>
  mutate(model = "long",
    .before = 1)

m2_beta <- m2_beta |>
  select(term, estimate, stars, starts_with("conf")) |>
  mutate(model = "short",
    .before = 1)

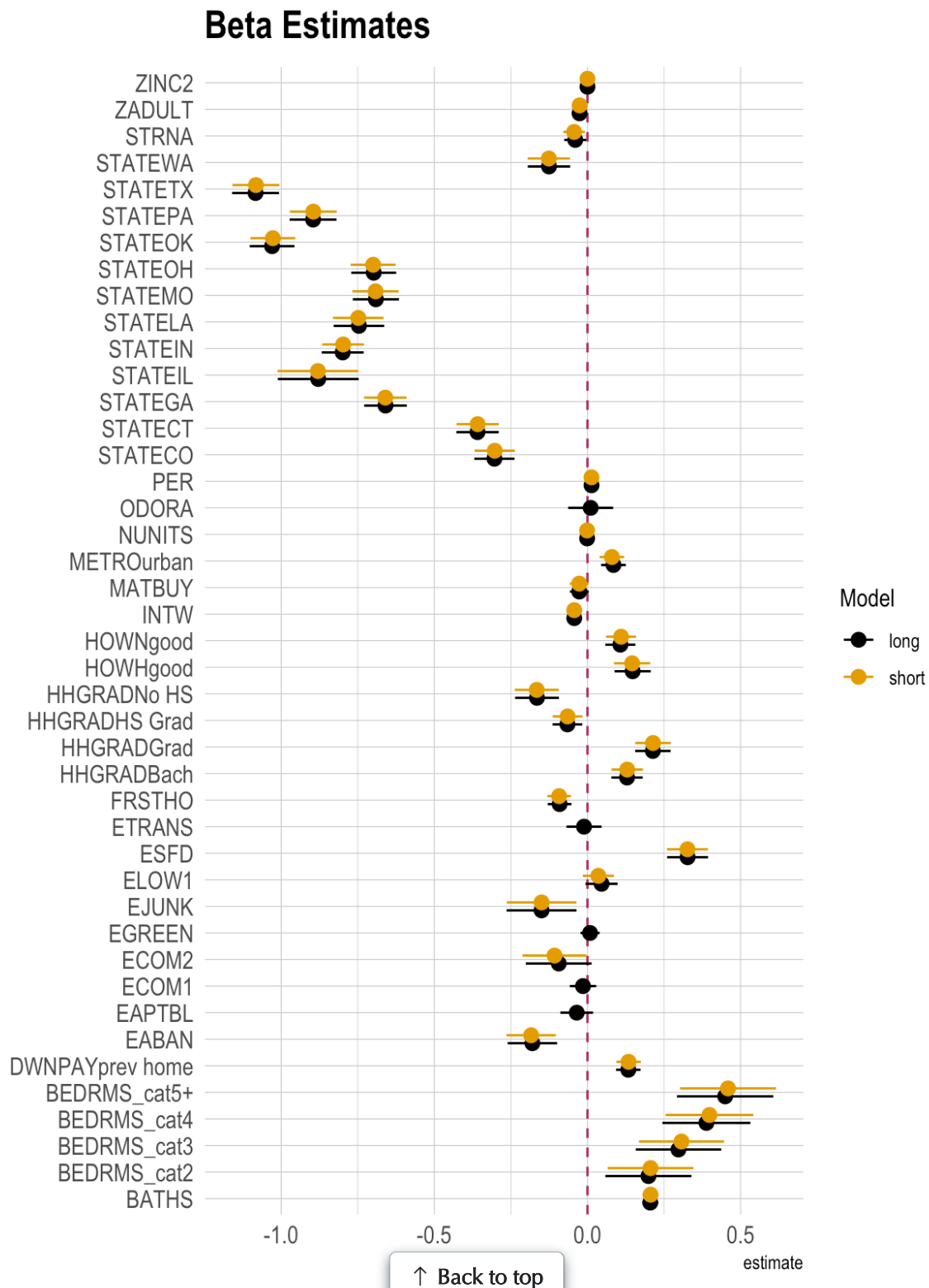
betas_linear <- m1_beta |>
  bind_rows(m2_beta) |>
  arrange(term, model)

betas_linear |>
  filter(!str_detect(term, "Intercept")) |>
  ggplot(
    aes(
      y = term,
      x = estimate,
      xmin = conf.low,
      xmax = conf.high,
      color = model
    )
  ) +
  geom_vline(xintercept = 0, color = "black", size = 2) +
  geom_pointrange(
    position = position_dodge(width = 0.6)
  ) +

```

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```
labs(  
  title = "Beta Estimates",  
  color = "Model",  
  y = ""  
)
```



- Beta estimates look almost same.

R^2 / AIC

```
m1_fit <- m1 |>
  glance()

m2_fit <- m2 |>
  glance()

m1_fit <- m1_fit |>
  mutate(model = "long",
         .before = 1)

m2_fit <- m2_fit |>
  mutate(model = "short",
         .before = 1)

m_fit <- m1_fit |>
  bind_rows(m2_fit)

m_fit |>
  paged_table()
```

model	r.squared	adj.r.squared	sigma
<chr>	<dbl>	<dbl>	<dbl>
long	0.3111422	0.3085987	0.8117424
short	0.3109503	0.3087030	0.8116812

2 rows | 1-4 of 13 columns

- In terms of AIC, the short model (`m1`) is a slightly better, more parsimonious model that fits the data better with less overfitting.

RMSE

```
dtest_resid <- dtest |>
  mutate(pred_1 = predict(m1, newdata = dtest),
         pred_2 = predict(m2, newdata = dtest),
         resid_1 = log(VALUE) - pred_1,
         resid_2 = log(VALUE) - pred_2
        ) |>
  select(VALUE,
         pred_1, pred_2,
         resid_1, resid_2)

RMSEs <- dtest_resid |>
  summarise(rmse_1 = sqrt(mean(resid_1^2)),
            rmse_2 = sqrt(mean(resid_2^2)),
            )
```

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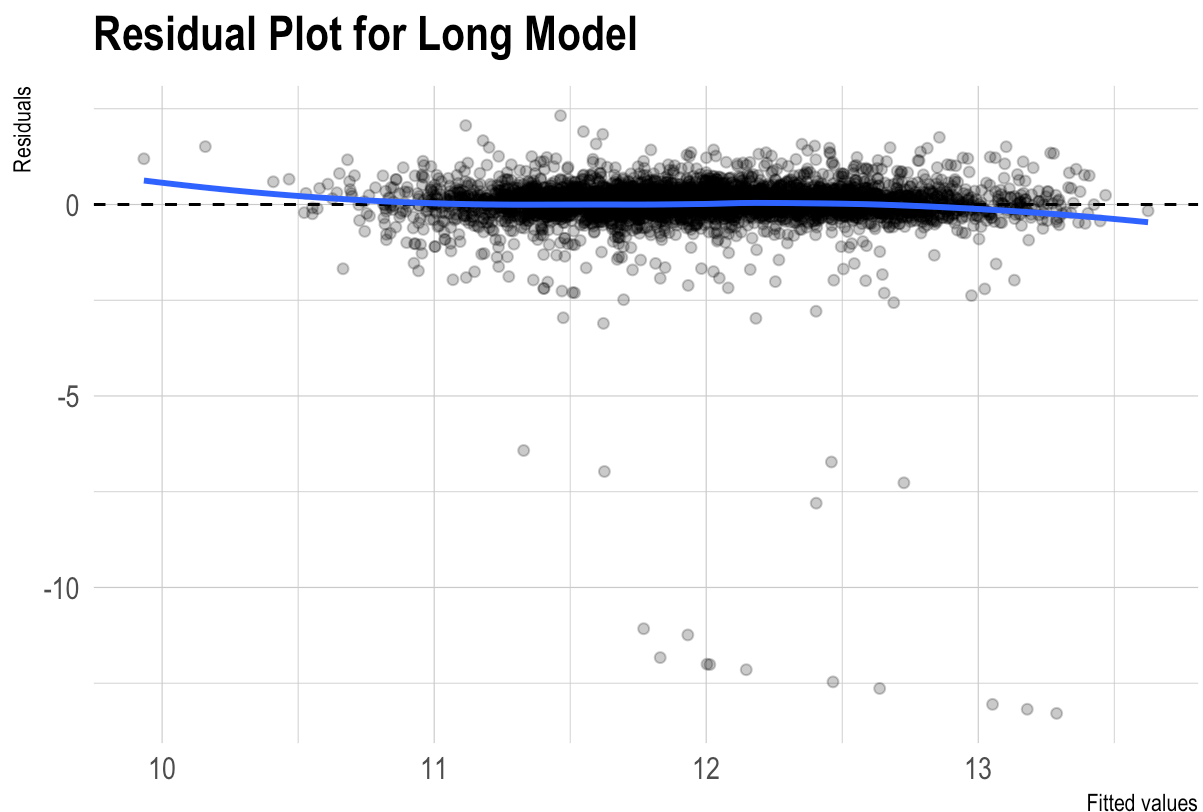
```
RMSEs |>
  paged_table()
```


	rmse_1	rmse_2
	<dbl>	<dbl>
	0.8338665	0.8341904
1 row		

- In terms of RMSE, the long model (`m1`) is a slightly better model, as a smaller RMSE reflects smaller differences between predicted and observed values.

Residual plots

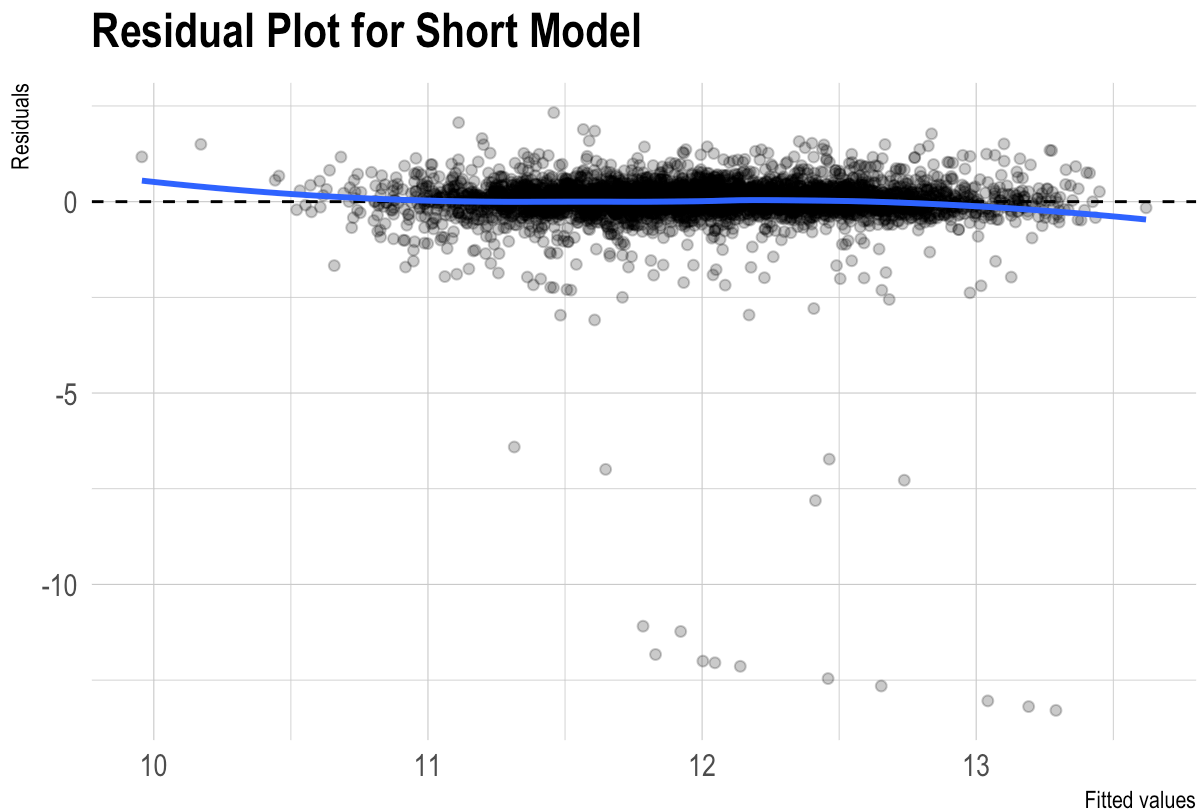
```
dtest_resid |>
  ggplot(aes(x = pred_1, y = resid_1)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_smooth(se = FALSE, method = "loess") +
  labs(
    x = "Fitted values",
    y = "Residuals",
    title = "Residual Plot for Long Model"
  )
)
```



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```
dtest_resid |>
  ggplot(aes(x = pred_2, y = resid_2)) +
  geom_point(alpha = 0.25) +
```

```
geom_hline(yintercept = 0, linetype = "dashed") +
geom_smooth(se = FALSE, method = "loess") +
labs(
  x = "Fitted values",
  y = "Residuals",
  title = "Residual Plot for Short Model"
)
```



- Both residual plots look almost identical.

Question 4

- Fit a **logistic regression model** with the following specifications:
 - **Outcome variable:** GT20DWN (indicating whether the buyer made a down payment of 20% or more)
 - **Predictors:** All available variables except **AMMORT** and **LPRICE**

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- The outcome variable is defined as:

$$GT20DWN = \begin{cases} 1 & \text{if } \frac{LPRICE - AMMORT}{LPRICE} > 0.2 \\ 0 & \text{otherwise} \end{cases}$$

- Analyze and interpret the following relationships:
- The association between **first-time homeownership** (FRSTHO) and the probability of making a 20%+ down payment.
- The association between **number of bedrooms** (BEDRMS) and the probability of making a 20%+ down payment.

Show answer

```
m1_logit <- glm(
  GT20DWN ~ .,
  data = dtrain |> select(-AMMORT, -LPRICE, -VALUE, -BEDRMS),
  family = binomial(link = "logit")
)

stargazer(
  m1_logit,
  type = "html",
  digits = 3,
  title = "Logistic regression (logit): GT20DWN"
)
```

Logistic regression (logit): GT20DWN

	Dependent variable:
	GT20DWN
STATECO	-0.178* (0.095)
STATECT	0.608*** (0.098)
STATEGA	-0.449*** (0.104)
STATEIL	0.169 (0.191)
STATEIN	-0.072 (0.101)
STATELA	0.281** (0.120)
STATEMO	0.179* (0.109)
STATEOH	0.497*** (0.105)

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STATEOK	-0.269** (0.111)
STATEPA	0.324*** (0.111)
STATETX	-0.100 (0.116)
STATEWA	0.091 (0.100)
METROurban	-0.017 (0.061)
ZINC2	0.00000 (0.00000)
HHGRADBach	0.166** (0.075)
HHGRADGrad	0.274*** (0.083)
HHGRADHS Grad	-0.056 (0.073)
HHGRADNo HS	-0.211* (0.115)
BATHS	0.325*** (0.038)
PER	-0.123*** (0.021)
ZADULT	0.012 (0.036)
NUNITS	0.003* (0.002)
EAPTBL	-0.023 (0.082)
ECOM1	-0.173*** (0.067)
ECOM2	-0.329* (0.185)
EGREEN	0.021 (0.046)
EJUNK	-0.015 (0.185)
ELow1	0.045 (0.077)
ESFD	-0.131 (0.097)
ETRANS	-0.099

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	(0.089)
EABAN	-0.087
	(0.133)
HOWHgood	-0.133
	(0.092)
HOWNgood	0.178**
	(0.077)
ODORA	0.051
	(0.113)
STRNA	-0.116**
	(0.055)
INTW	-0.094***
	(0.016)
MATBUY	0.297***
	(0.045)
DWNPAYprev home	0.770***
	(0.055)
FRSTHO	-0.403***
	(0.059)
BEDRMS_cat2	0.425*
	(0.226)
BEDRMS_cat3	0.229
	(0.225)
BEDRMS_cat4	0.371
	(0.230)
BEDRMS_cat5+	0.254
	(0.248)
Constant	-1.240***
	(0.295)

Observations	11,690
--------------	--------

Log Likelihood	-6,417.980
----------------	------------

Akaike Inf. Crit.	12,923.960
-------------------	------------

Note: $p < 0.1$; **$p < 0.05$** ; $p < 0.01$

```
m1_logit_betas <- m1_logit |>
  tidy(conf.int = T) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "****", # smallest p-values first
      p.value < .05 ~ "***",
      p.value < .1 ~ "**",
      TRUE ~ ""
    ),
    .after = estimate
  )
```

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```
m1_logit_betas |>
  paged_table()
```

term	estimate	stars
<chr>	<dbl>	<chr>
(Intercept)	-1.239906e+00	***
STATECO	-1.775604e-01	*
STATECT	6.081423e-01	***
STATEGA	-4.487022e-01	***
STATEIL	1.694266e-01	
STATEIN	-7.154044e-02	
STATELA	2.810709e-01	**
STATEMO	1.790601e-01	*
STATEOH	4.971107e-01	***
STATEOK	-2.694479e-01	**

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AME

```
sum_me <- summary(
  margins(m1_logit,
    variables = c("FRSTHO",
                  "BEDRMS_cat"))
)

sum_me
```

factor	AME	SE	z	p	lower	upper
BEDRMS_cat2	0.0762	0.0377	2.0221	0.0432	0.0023	0.1501
BEDRMS_cat3	0.0397	0.0373	1.0652	0.2868	-0.0334	0.1128
BEDRMS_cat4	0.0658	0.0384	1.7147	0.0864	-0.0094	0.1411
BEDRMS_cat5+	0.0442	0.0417	1.0610	0.2887	-0.0375	0.1259
FRSTHO	-0.0742	0.0109	-6.8261	0.0000	-0.0955	-0.0529

- The relationship between **number of bedrooms (BEDRMS)** and the probability of making a **20%+ down payment**:
 - On average, homes with **two bedrooms** are estimated to have a **7.62 percentage point higher probability** of a **20% or higher down payment** than homes with **one bedroom or no bedroom**, holding other covariates constant.
 - Intuition:** Relative to very small homes, two-bedroom homes may be more commonly purchased by households with greater financial resources, making a larger down payment more attainable.
 - On average, homes with **four bedrooms** are estimated to have a **6.58 percentage point higher probability** of a **20% or higher down payment** than homes with **one bedroom or no bedroom**, holding other covariates constant.
 - Intuition:** Four-bedroom homes may be more commonly purchased by larger or more financially established households with enough savings to make a large down payment upfront.
 - By contrast, the estimated relationships for homes with **three bedrooms** and **five or more bedrooms** are **not statistically significant**.

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- **Intuition:** Their estimated effects are too imprecise to rule out the possibility that the true differences from the reference group are small or zero.
- The association between **first-time homeownership (FRSTHO)** and the probability of making a 20%+ down payment:
 - On average, first homes have an estimated **7.61 percentage point lower probability** of making a **20% or higher down payment** than non-first homes, holding other covariates constant.
 - **Intuition:** First-time buyers often have less accumulated savings and fewer housing-related assets (for example, they cannot use equity from selling a previous home), so they are less likely to be able to put down 20% or more.

Question 5

- Refit the **logistic regression model**, adding interaction terms:
 - **Predictors:** all previously included predictors in Question 4 **plus** the interaction between **FRSTHO** and **BEDRMS**
- Interpret how the relationship between **BEDRMS** and the probability of a 20%+ down payment varies depending on whether the buyer is a first-time homeowner (**FRSTHO**).

Show answer

```
m2_logit <- glm(
  GT20DWN ~ . + BEDRMS_cat:FRSTHO,
  data = dtrain |>
    select(-AMMORT, -LPRICE, -VALUE, -BEDRMS),
  family = binomial(link = "logit")
)

stargazer(
  m2_logit,
  type = "html",
  digits = 3,
  title = "Logistic regression (logit): GT20DWN with an Interaction"
)
```

Logistic regression (logit): GT20DWN with an Interaction

	<i>Dependent variable:</i>
	GT20DWN
STATECO	-0.178* (0.095)
STATECT	0.613*** (0.098)
STATEGA	-0.453*** (0.104)

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STATEIL	0.174 (0.191)
STATEIN	-0.073 (0.101)
STATELA	0.280** (0.120)
STATEMO	0.180* (0.109)
STATEOH	0.498*** (0.105)
STATEOK	-0.267** (0.111)
STATEPA	0.324*** (0.111)
STATETX	-0.104 (0.116)
STATEWA	0.095 (0.100)
METROurban	-0.013 (0.061)
ZINC2	0.00000 (0.00000)
HHGRADBach	0.165** (0.075)
HHGRADGrad	0.272*** (0.083)
HHGRADHS Grad	-0.056 (0.073)
HHGRADNo HS	-0.213* (0.115)
BATHS	0.323*** (0.038)
PER	-0.125*** (0.021)
ZADULT	0.017 (0.036)
NUNITS	0.003 (0.002)
EAPTBL	-0.020 (0.082)
ECOM1	-0.172** (0.067)
ECOM2	-0.333*

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	(0.185)
EGREEN	0.020
	(0.046)
EJUNK	-0.013
	(0.185)
ELOW1	0.044
	(0.077)
ESFD	-0.135
	(0.097)
ETRANS	-0.099
	(0.089)
EABAN	-0.089
	(0.133)
HOWHgood	-0.135
	(0.092)
HOWNgood	0.179**
	(0.077)
ODORA	0.053
	(0.113)
STRNA	-0.114**
	(0.055)
INTW	-0.094***
	(0.016)
MATBUY	0.297***
	(0.045)
DWNPAYprev home	0.769***
	(0.056)
FRSTHO	-0.246
	(0.427)
BEDRMS_cat2	0.538*
	(0.320)
BEDRMS_cat3	0.291
	(0.316)
BEDRMS_cat4	0.461
	(0.319)
BEDRMS_cat5+	0.421
	(0.333)
FRSTHO:BEDRMS_cat2	-0.224
	(0.441)
FRSTHO:BEDRMS_cat3	-0.097
	(0.430)
FRSTHO:BEDRMS_cat4	-0.174
	(0.437)

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FRSTHO:BEDRMS_cat5+	-0.713
	(0.499)
Constant	-1.317***
	(0.366)
Observations	11,690
Log Likelihood	-6,414.807
Akaike Inf. Crit.	12,925.610

Note: $p<0.1$; $p<0.05$; $p<0.01$

```
m2_logit_betas <- m2_logit |>
  tidy(conf.int = T) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "****", # smallest p-values first
      p.value < .05 ~ "***",
      p.value < .1 ~ "**",
      TRUE ~ ""
    ),
    .after = estimate
  )

m2_logit_betas |>
  paged_table()
```

term	estimate	stars
<chr>	<dbl>	<chr>
(Intercept)	-1.316849e+00	***
STATECO	-1.778993e-01	*
STATECT	6.126788e-01	***
STATEGA	-4.534357e-01	***
STATEIL	1.742297e-01	
STATEIN	-7.340710e-02	
STATELA	2.795359e-01	**
STATEMO	1.799327e-01	*
STATEOH	4.976258e-01	***
STATEOK	-2.667889e-01	**

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AME

```
sum_me2 <- summary(
  margins(m2_logit,
    variables = c("FRSTHO", "BEDRMS_cat"),
    at = list(
      BEDRMS_cat = levels(dtrain$B
    )
  )
)
```

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```
sum_me2 |>
  paged_table()
```

	factor <chr>	BEDRMS_cat <fct>	AME <dbl>
1	BEDRMS_cat2	1 or fewer	0.08328061
2	BEDRMS_cat2	2	0.08328061
3	BEDRMS_cat2	3	0.08328061
4	BEDRMS_cat2	4	0.08328061
5	BEDRMS_cat2	5+	0.08328061
6	BEDRMS_cat3	1 or fewer	0.04498269
7	BEDRMS_cat3	2	0.04498269
8	BEDRMS_cat3	3	0.04498269
9	BEDRMS_cat3	4	0.04498269
10	BEDRMS_cat3	5+	0.04498269

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`at = option in margins()`

- In this `margins()` code, the `at =` option tells R to compute marginal effects **at specific representative values** of a predictor rather than averaging over the observed distribution of that predictor.
- Here, `at = list(BEDRMS_cat = levels(dtrain$BEDRMS_cat))` asks `margins()` to evaluate the marginal effects **at each bedroom category**: for example, **1 or fewer, 2, 3, 4, and 5+** bedrooms.
- This is similar in spirit to the idea of a **marginal effect at a representative value (MER)** that we discussed in class. The difference is that, instead of evaluating the effect at the **mean** of a numeric variable, we are evaluating it at each **representative category** of a factor variable.
- This is especially useful when the model includes an interaction, because the marginal effect of one variable may differ depending on the value or category of another variable.
- In this example, the `at =` option lets us see how the estimated effect of `FRSTHO` changes across bedroom categories, rather than reporting only one overall average marginal effect.
- So, you can think of `at =` as asking: **“What is the marginal effect when we fix a predictor at a particular representative value or category?”**
- In our case, it answers: **“What is the marginal effect of first-time homebuyer status for homes with 1 or fewer bedrooms, 2 bedrooms, 3 bedrooms, 4 bedrooms, and 5+ bedrooms?”**

- The relationship between **number of bedrooms (BEDRMS)** and the probability of making a **down payment greater than 20%**:
 - Relative to homes with **one bedroom or no bedroom** (the reference group), homes with **two bedrooms** have an estimated **8.33 percentage point higher probability** of making a **down payment greater than 20%**, on average, holding other covariates constant.
 - This estimate is **statistically significant** at the 5% level ($p = 0.038$).

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- **Intuition:** Compared with very small homes, two-bedroom homes may be more likely to be purchased by households with somewhat greater financial stability or savings, making a larger down payment more feasible.
- Relative to homes with **one bedroom or no bedroom**, homes with **three bedrooms** have an estimated **4.50 percentage point higher probability** of making a **down payment greater than 20%**, on average, holding other covariates constant.
 - However, this estimate is **not statistically significant** ($p = 0.256$).
 - **Intuition:** Although the estimated association is positive, the evidence is not strong enough to conclude that three-bedroom homes differ meaningfully from the reference group after accounting for the other predictors in the model.
- Relative to homes with **one bedroom or no bedroom**, homes with **four bedrooms** have an estimated **7.19 percentage point higher probability** of making a **down payment greater than 20%**, on average, holding other covariates constant.
 - This estimate is **marginally significant** at the 10% level ($p = 0.077$), but not at the 5% level.
 - **Intuition:** Four-bedroom homes may be more likely to be purchased by larger or more established households with greater accumulated savings, but the evidence here is only moderately strong.
- Relative to homes with **one bedroom or no bedroom**, homes with **five or more bedrooms** have an estimated **3.65 percentage point higher probability** of making a **down payment greater than 20%**, on average, holding other covariates constant.
 - This estimate is **not statistically significant** ($p = 0.402$).
 - **Intuition:** While the point estimate is positive, it is too imprecise to conclude that homes with five or more bedrooms are meaningfully different from the reference group.
- The relationship between **first-time homebuyer status (FRSTHO)** and the probability of making a **down payment greater than 20%**, by bedroom category:
 - Among homes with **one bedroom or no bedroom**, first homes have an estimated **4.05 percentage point lower probability** of making a **down payment greater than 20%** than non-first homes, on average, holding other covariates constant.
 - This estimate is **not statistically significant** ($p = 0.567$).
 - **Intuition:** For very small homes, the data do not provide strong evidence that first-time buyers differ from non-first-time buyers in down payment behavior.
 - Among homes with **two bedrooms**, first homes have an estimated **9.00 percentage point lower probability** of making a **down payment greater than 20%** than non-first homes, on average, holding other covariates constant.
 - This estimate is **statistically significant** at the 1% level ($p < 0.001$).
 - **Intuition:** First-time buyers of two-bedroom homes may have less accumulated wealth or home equity to put toward a large down payment than repeat buyers.
 - Among homes with **three bedrooms**, first homes have an estimated **6.22 percentage point lower probability** of making a **down payment greater than 20%** than non-first homes, on average, holding other covariates constant.
 - This estimate is **statistically significant** at the 1% level ($p < 0.001$).
 - **Intuition:** This pattern is consistent with the idea that repeat buyers can often use proceeds or equity from a previous home sale to make a larger down payment.
 - Among homes with **four bedrooms**, first homes have an estimated **7.94 percentage point lower probability** of making a **down payment greater than 20%** than non-first homes, on average, holding other covariates constant.
 - This estimate is **statistically significant** at the 1% level ($p < 0.001$).
 - **Intuition:** For larger homes, first-time buyers may face greater financing constraints because these homes generally require more upfront cash.

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- Among homes with **five or more bedrooms**, first homes have an estimated **16.13 percentage point lower probability** of making a **down payment greater than 20%** than non-first homes, on average, holding other covariates constant.
 - This estimate is **statistically significant** at the 1% level ($p < 0.001$).
 - **Intuition:** The gap is especially large for the biggest homes, suggesting that repeat buyers may have a particularly strong financial advantage in making large down payments in this segment of the housing market.

Question 6

- Fit separate **logistic regression models** (with the same model specification as in Question 4) for two subsets of home data:
 1. Homes worth $\text{VALUE} \geq 175k$.
 2. Homes worth $\text{VALUE} < 175k$.
- Compare residual deviance, *RMSE*, and classification performance between the two models.

Show answer

Subsetting Data

```
dtrain_over_175k <- dtrain |>
  filter(VALUE >= 175000)
dtrain_under_175k <- dtrain |>
  filter(VALUE < 175000)

dtest_over_175k <- dtest |>
  filter(VALUE >= 175000)
dtest_under_175k <- dtest |>
  filter(VALUE < 175000)
```

Fitting the Models

```
m1_logit_over_175k <- glm(
  GT20DWN ~ .,
  data = dtrain_over_175k |> select(-AMMORT, -LPRICE, -VALUE, -BEDRMS),
  family = binomial(link = "logit")
)

stargazer(
  m1_logit_over_175k,
  type = "html",
  digits = 3,
  title = "Logistic regression (logit): GT20DWN (VALUE over 175k)"
)
```

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Logistic regression (logit): GT20DWN
(VALUE over 175k)

	<i>Dependent variable:</i>
	GT20DWN
STATECO	-0.129 (0.101)
STATECT	0.623*** (0.111)
STATEGA	-0.369*** (0.125)
STATEIL	0.116 (0.323)
STATEIN	0.221 (0.143)
STATELA	0.498*** (0.179)
STATEMO	0.481*** (0.141)
STATEOH	0.529*** (0.145)
STATEOK	0.082 (0.186)
STATEPA	0.549*** (0.174)
STATETX	0.163 (0.208)
STATEWA	0.088 (0.106)
METROurban	0.018 (0.089)
ZINC2	0.00000 (0.00000)
HHGRADBach	0.211** (0.103)
HHGRADGrad	0.287*** (0.110)
HHGRADHS Grad	0.007 (0.105)
HHGRADNo HS	-0.361* (0.192)
BATHS	0.359*** (0.047)

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PER	-0.103*** (0.027)
ZADULT	-0.023 (0.048)
NUNITS	0.003 (0.006)
EAPTBL	0.063 (0.126)
ECOM1	-0.019 (0.096)
ECOM2	0.047 (0.318)
EGREEN	0.077 (0.060)
EJUNK	-0.081 (0.309)
ELOW1	0.077 (0.108)
ESFD	-0.241 (0.158)
ETRANS	-0.191 (0.137)
EABAN	-0.460* (0.261)
HOWHgood	-0.217 (0.145)
HOWNgood	0.466*** (0.120)
ODORA	0.089 (0.168)
STRNA	-0.185** (0.077)
INTW	-0.076*** (0.025)
MATBUY	0.535*** (0.060)
DWNPAYprev home	0.752*** (0.072)
FRSTHO	-0.367*** (0.087)
BEDRMS_cat2	0.450 (0.454)
BEDRMS_cat3	0.329

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	(0.450)
BEDRMS_cat4	0.451
	(0.454)
BEDRMS_cat5+	0.226
	(0.466)
Constant	-1.736***
	(0.535)
Observations	5,903
Log Likelihood	-3,508.690
Akaike Inf. Crit.	7,105.381

Note: $p<0.1$; **$p<0.05$** ; $p<0.01$

```
m1_logit_betas_over_175k <- m1_logit_over_175k |>
  tidy(conf.int = T) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "****", # smallest p-values first
      p.value < .05 ~ "***",
      p.value < .1 ~ "**",
      TRUE ~ ""
    ),
    .after = estimate
  )
```

```
m1_logit_under_175k <- glm(
  GT20DWN ~ .,
  data = dtrain_under_175k |> select(-AMMORT, -LPRICE, -VALUE, -BEDRMS),
  family = binomial(link = "logit")
)

stargazer(
  m1_logit_under_175k,
  type = "html",
  digits = 3,
  title = "Logistic regression (logit): GT20DWN (VALUE under 175k)"
)
```

Logistic regression (logit): GT20DWN
(VALUE under 175k)

	<i>Dependent variable:</i>
	GT20DWN
STATECO	-0.888***
	(0.333)
STATECT	0.106
	(0.301)
STATEGA	-0.924***
	(0.303)

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STATEIL	-0.263 (0.349)
STATEIN	-0.698** (0.292)
STATELA	-0.246 (0.303)
STATEMO	-0.570* (0.302)
STATEOH	0.005 (0.294)
STATEOK	-0.790*** (0.295)
STATEPA	-0.274 (0.296)
STATETX	-0.535* (0.298)
STATEWA	-0.196 (0.336)
METROurban	-0.068 (0.088)
ZINC2	0.00000 (0.00000)
HHGRADBach	0.052 (0.115)
HHGRADGrad	0.149 (0.138)
HHGRADHS Grad	-0.103 (0.103)
HHGRADNo HS	-0.135 (0.148)
BATHS	0.153** (0.068)
PER	-0.163*** (0.035)
ZADULT	0.075 (0.058)
NUNITS	0.003 (0.002)
EAPTBL	-0.077 (0.111)
ECOM1	-0.324*** (0.095)
ECOM2	-0.534**

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	(0.239)
EGREEN	-0.096
	(0.073)
EJUNK	-0.042
	(0.232)
ELOW1	0.043
	(0.114)
ESFD	-0.072
	(0.128)
ETRANS	0.020
	(0.119)
EABAN	0.050
	(0.155)
HOWHgood	-0.056
	(0.120)
HOWNgood	-0.061
	(0.102)
ODORA	0.003
	(0.155)
STRNA	-0.041
	(0.079)
INTW	-0.087***
	(0.021)
MATBUY	-0.049
	(0.070)
DWNPAYprev home	0.714***
	(0.091)
FRSTHO	-0.378***
	(0.083)
BEDRMS_cat2	0.427
	(0.261)
BEDRMS_cat3	0.199
	(0.264)
BEDRMS_cat4	0.234
	(0.279)
BEDRMS_cat5+	0.375
	(0.352)
Constant	-0.124
	(0.440)

Observations	5,787
Log Likelihood	-2,836.965
Akaike Inf. Crit.	5,761.931

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Note: $p < 0.1$; $p < 0.05$; $p < 0.01$

```
m1_logit_betas_under_175k <- m1_logit_under_175k |>
  tidy(conf.int = T) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "****", # smallest p-values first
      p.value < .05 ~ "***",
      p.value < .1 ~ "**",
      TRUE ~ ""
    ),
    .after = estimate
  )
```

Betas

```
m1_logit_betas_over_175k <- m1_logit_betas_over_175k |>
  select(term, estimate, stars, starts_with("conf")) |>
  mutate(model = "VALUE over 175k",
    .before = 1)

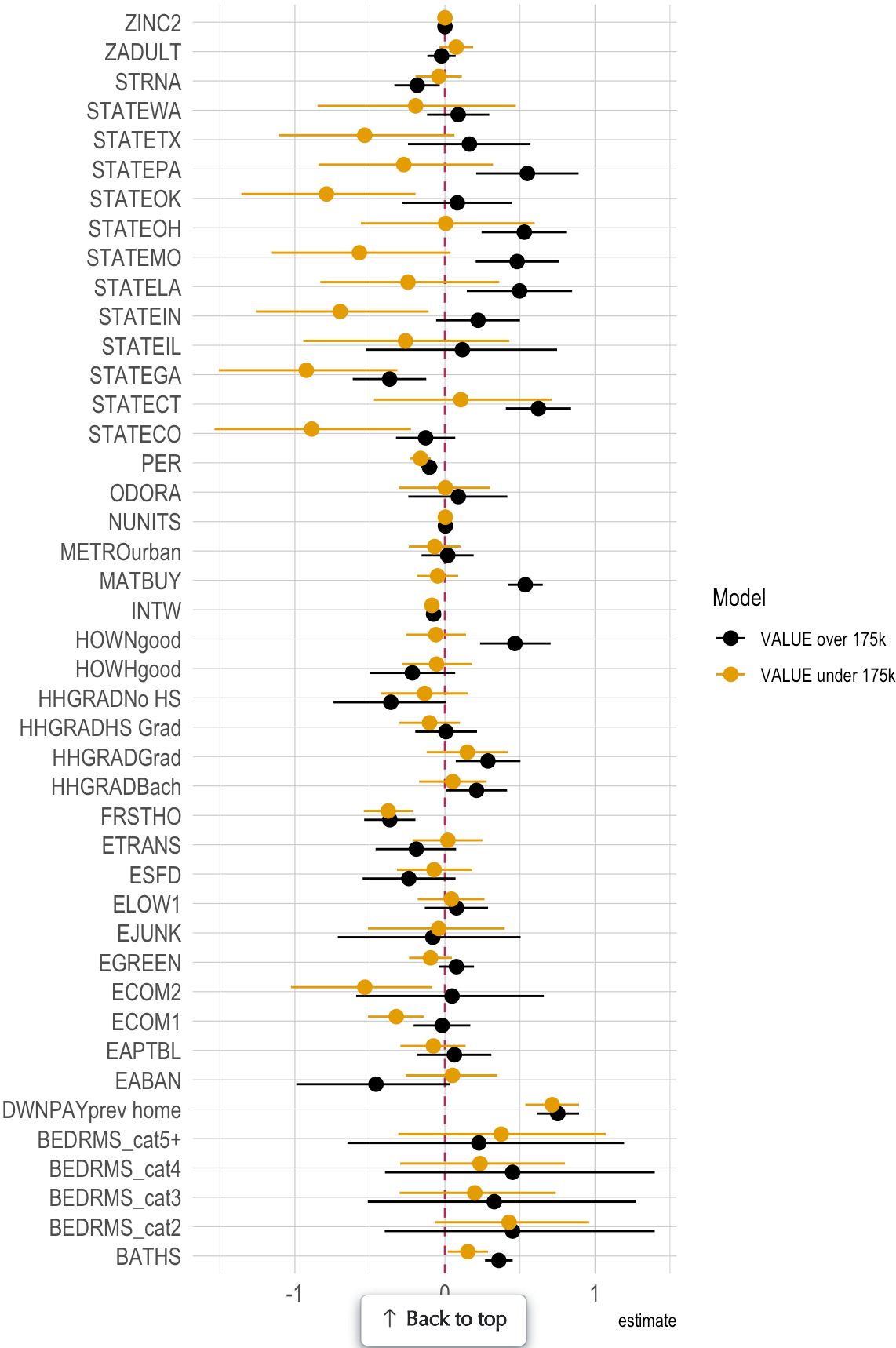
m1_logit_betas_under_175k <- m1_logit_betas_under_175k |>
  select(term, estimate, stars, starts_with("conf")) |>
  mutate(model = "VALUE under 175k",
    .before = 1)

betas_logit_175k <- m1_logit_betas_over_175k |>
  bind_rows(m1_logit_betas_under_175k) |>
  arrange(term, model)

betas_logit_175k |>
  filter(!str_detect(term, "Intercept")) |>
  ggplot(
    aes(
      y = term,
      x = estimate,
      xmin = conf.low,
      xmax = conf.high,
      color = model
    )
  ) +
  geom_vline(xintercept = 0, color = "maroon", linetype = 2) +
  geom_pointrange(
    position = position_dodge(width = 0.6)
  ) +
  labs(
    title = "Beta Estimates",
    color = "Model",
    y = ""
  )
```

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Beta Estimates



AMEs of FRSTHO and BEDRMS_cat

```
sum_me_over_175k <- summary(
  margins(m1_logit_under_175k,
    variables = c("FRSTHO", "BEDRMS_cat")
  )
) |>
mutate(model = "VALUE over 175k",
  .before = 1)

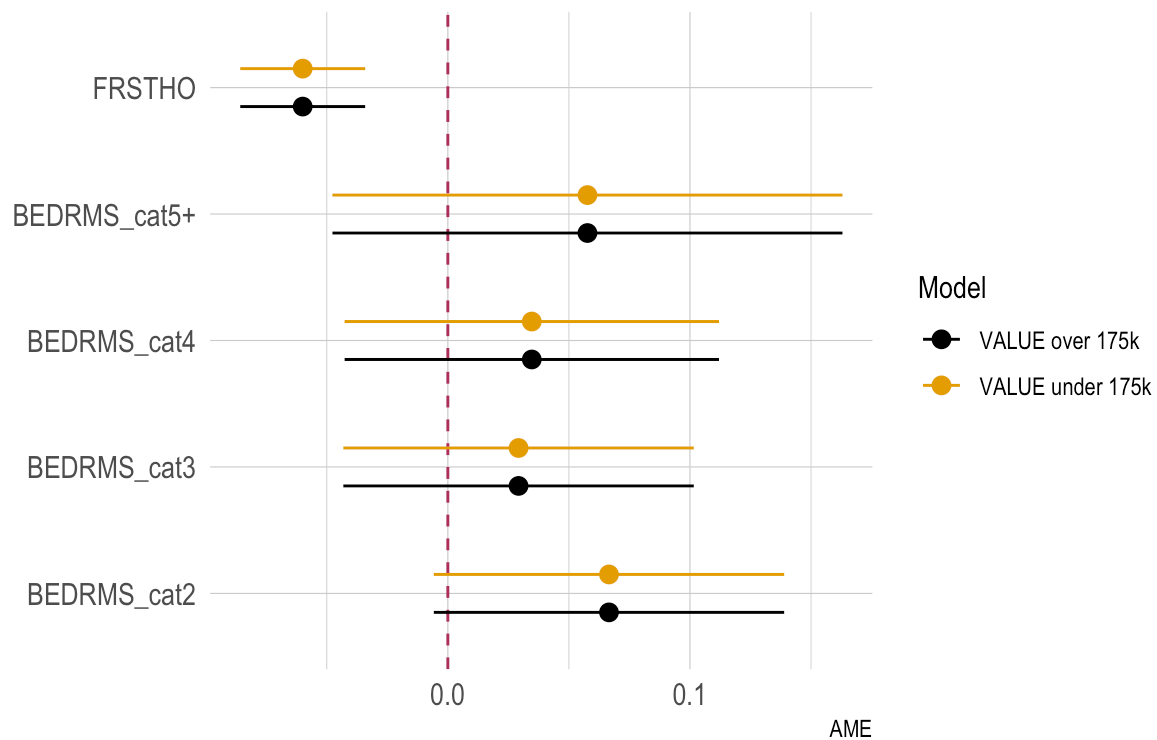
sum_me_under_175k <- summary(
  margins(m1_logit_under_175k,
    variables = c("FRSTHO", "BEDRMS_cat")
  )
) |>
mutate(model = "VALUE under 175k",
  .before = 1)

AME_logit_175k <- sum_me_over_175k |>
bind_rows(sum_me_under_175k) |>
arrange(factor, model)

AME_logit_175k |>
ggplot(
  aes(
    y = factor,
    x = AME,
    xmin = lower,
    xmax = upper,
    color = model
  )
) +
geom_vline(xintercept = 0, color = "maroon", linetype = 2) +
geom_pointrange(
  position = position_dodge(width = 0.6)
) +
labs(
  title = "AME Estimates",
  color = "Model",
  y = ""
)
```

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AME Estimates



Residual Deviance

```
m1_logit_over_175k |>
  glance() # focus on the `deviance` column
```

A tibble: 1 × 8

	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	7776.	5902	-3509.	7105.	7399.	7017.	5859	5903

```
m1_logit_under_175k |>
  glance() # focus on the `deviance` column
```

A tibble: 1 × 8

	null.deviance	df.null	logLik	AIC	BIC	deviance	df.residual	nobs
	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	6090.	5786	-2837.	5762.	6055.	5674.	5743	5787

RMSE

```
dtest_resid_over_175k <- dtest_over_175k
mutate(.fitted = predict(m1_logit_over_175k,
  newdata = dtest_over_175k,
  type = "response"),
```

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```

      .resid = GT20DWN - .fitted
    ) |>
  select(GT20DWN,
         .fitted, .resid)

dtest_resid_under_175k <- dtest_under_175k |>
  mutate(.fitted = predict(m1_logit_under_175k,
                          newdata = dtest_under_175k,
                          type = "response"),
         .resid = GT20DWN - .fitted
  ) |>
  select(GT20DWN,
         .fitted, .resid)

RMSE_over_175k <- dtest_resid_over_175k |>
  summarise(rmse = sqrt(mean(.resid^2))
  )

RMSE_under_175k <- dtest_resid_under_175k |>
  summarise(rmse = sqrt(mean(.resid^2))
  )

RMSE_over_175k |>
  paged_table()

```

rmse

<dbl>

0.4492962

1 row

```

RMSE_under_175k |>
  paged_table()

```

rmse

<dbl>

0.4021622

1 row

Classification Performance with the Same Threshold

Double Density Plot

```

threshold_over_175k <- .5

m1_logit_over_175k |>
  augment(type.predict = "response") |>
  ggplot(aes(x = .fitted,
            fill = factor(GT20DWN))) +
  geom_density(alpha = 0.35) +
  geom_vline(xintercept = threshold_over_175k, linetype = "dashed") +
  labs(

```

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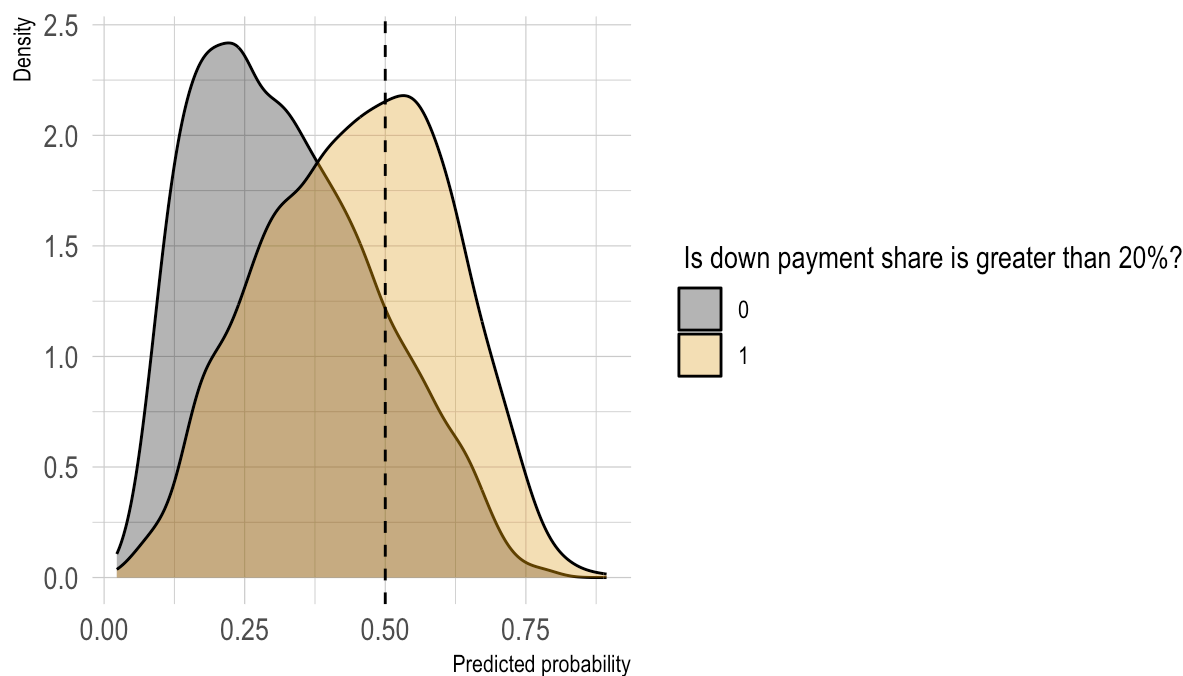
```

x = "Predicted probability",
y = "Density",
fill = " Is down payment share is greater than 20%?",
title = "Training set: predicted probabilities by actual class",
subtitle = "VALUE over 175k"
)

```

Training set: predicted probabilities by actual class

VALUE over 175k



```

threshold_under_175k <- .5

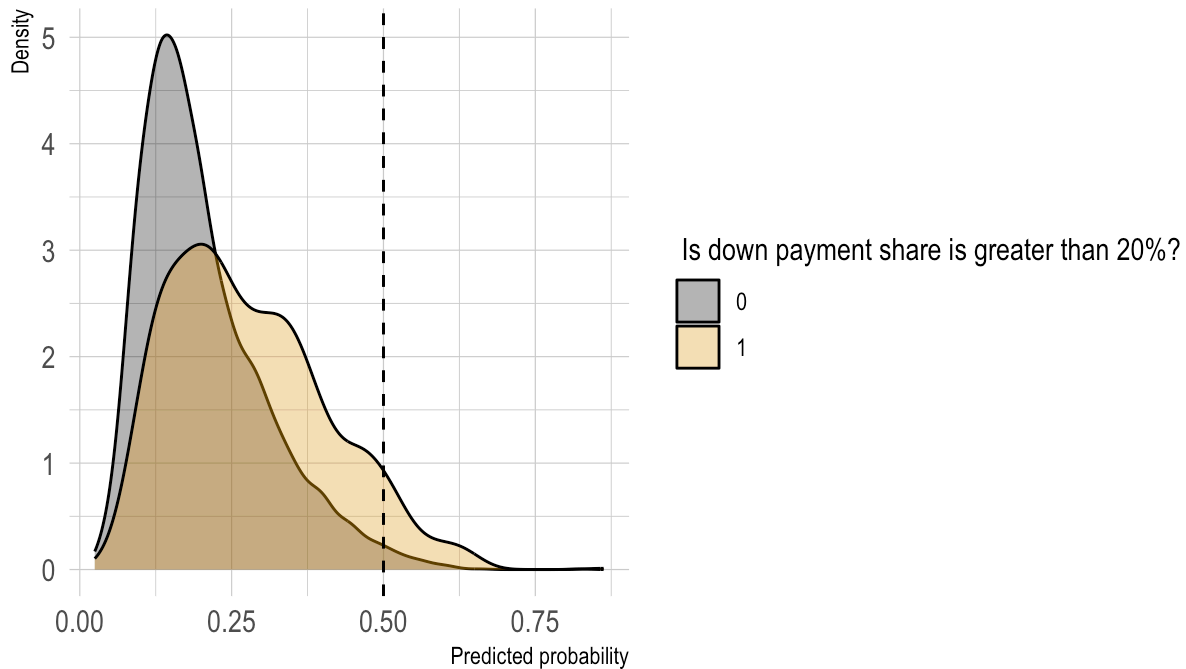
m1_logit_under_175k |>
  augment(type.predict = "response") |>
  ggplot(aes(x = .fitted,
             fill = factor(GT20DOWN))) +
  geom_density(alpha = 0.35) +
  geom_vline(xintercept = threshold_under_175k, linetype = "dashed") +
  labs(
    x = "Predicted probability",
    y = "Density",
    fill = " Is down payment share is greater than 20%",
    title = "Training set: predicted probabilities by actual class",
    subtitle = "VALUE under 175k"
  )
)

```

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Training set: predicted probabilities by actual class

VALUE under 175k



Confusion Matrix

```
df_cm_over_175k <- m1_logit_over_175k |>
  augment(newdata = dtest_over_175k,
    type.predict = "response") |>
  mutate(
    actual = factor(GT20DWN,
      levels = c(0, 1),
      labels = c("DWN <= 20%", "DWN > 20%")),
    pred = factor(if_else(.fitted > threshold_over_175k, 1, 0),
      levels = c(0, 1),
      labels = c("DWN <= 20%", "DWN > 20%")),
  )

conf_mat_over_175k <- table(truth = df_cm_over_175k$actual,
  prediction = df_cm_over_175k$pred)

conf_mat_over_175k
```

	prediction	
truth	DWN <= 20%	DWN > 20%
DWN <= 20%	1029	185
DWN > 20%	409	305

```
df_cm_under_175k <- m1_logit_under_175k
  augment(newdata = dtest_under_175k,
    type.predict = "response") |>
  mutate(
```

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```

actual = factor(GT20DWN,
                 levels = c(0, 1),
                 labels = c("DWN <= 20%", "DWN > 20%")),
pred    = factor(if_else(.fitted > threshold_under_175k, 1, 0),
                 levels = c(0, 1),
                 labels = c("DWN <= 20%", "DWN > 20%")),
)

```

```

conf_mat_under_175k <- table(truth = df_cm_under_175k$actual,
                             prediction = df_cm_under_175k$pred)
conf_mat_under_175k

```

	prediction	
truth	DWN <= 20%	DWN > 20%
DWN <= 20%	1507	19
DWN > 20%	404	17

Performance Metrics

```

base_rate_over_175k <- mean(dtest_over_175k$GT20DWN)

accuracy_over_175k <- (conf_mat_over_175k[1,1] + conf_mat_over_175k[2,2]) / sum(conf_mat_over_175k[,1])
precision_over_175k <- conf_mat_over_175k[2,2] / sum(conf_mat_over_175k[,2])
recall_over_175k <- conf_mat_over_175k[2,2] / sum(conf_mat_over_175k[2,])
specificity_over_175k <- conf_mat_over_175k[1,1] / sum(conf_mat_over_175k[1,])
enrichment_over_175k <- precision_over_175k / base_rate_over_175k

```

```

df_class_metric_over_175k <-
  data.frame(
    metric = c("Base rate",
               "Accuracy",
               "Precision",
               "Recall",
               "Specificity",
               "Enrichment"),
    value = c(base_rate_over_175k,
              accuracy_over_175k,
              precision_over_175k,
              recall_over_175k,
              specificity_over_175k,
              enrichment_over_175k)
  )

```

```

df_class_metric_over_175k |>
  mutate(value = round(value, 4)) |>
  rmarkdown::paged_table()

```

metric	value
<chr>	<dbl>
Base rate	0.3703
Accuracy	0.6919
Precision	0.6224

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metric	value
<chr>	<dbl>
Recall	0.4272
Specificity	0.8476
Enrichment	1.6808
6 rows	

```
base_rate_under_175k <- mean(dtest_under_175k$GT20DWN)

accuracy_under_175k <- (conf_mat_under_175k[1,1] + conf_mat_under_175k[2,2]) / sum(conf_mat_under_175k[,1])
precision_under_175k <- conf_mat_under_175k[2,2] / sum(conf_mat_under_175k[,2])
recall_under_175k <- conf_mat_under_175k[2,2] / sum(conf_mat_under_175k[2,])
specificity_under_175k <- conf_mat_under_175k[1,1] / sum(conf_mat_under_175k[1,])
enrichment_under_175k <- precision_under_175k / base_rate_under_175k

df_class_metric_under_175k <-
  data.frame(
    metric = c("Base rate",
               "Accuracy",
               "Precision",
               "Recall",
               "Specificity",
               "Enrichment"),
    value = c(base_rate_under_175k,
               accuracy_under_175k,
               precision_under_175k,
               recall_under_175k,
               specificity_under_175k,
               enrichment_under_175k)
  )

df_class_metric_under_175k |>
  mutate(value = round(value, 4)) |>
  rmarkdown::paged_table()
```

metric	value
<chr>	<dbl>
Base rate	0.2162
Accuracy	0.7827
Precision	0.4722
Recall	0.7245
Specificity	0.9875
Enrichment	2.1839
6 rows	

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Why a Common Classification Threshold Outperformed Separate Cutoffs

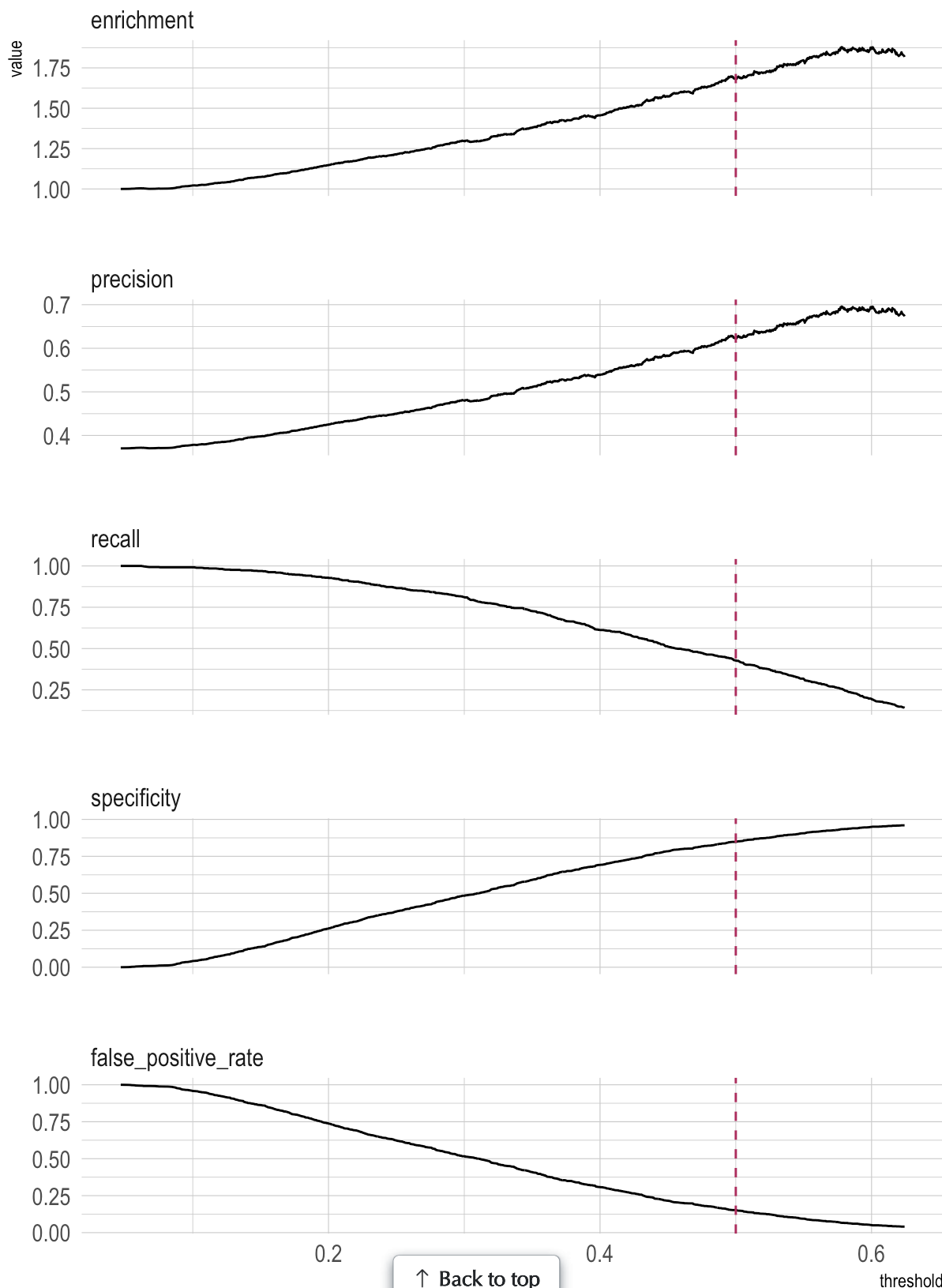
- I considered using different thresholds for the two subgroup models based on the double-density plots of predicted probabilities. However, the classification results were better when the same threshold was used for both groups.
- Intuitively, the density plots help visualize overlap between the two classes, but they do not directly choose the threshold that maximizes predictive performance. As a result, a visually appealing subgroup-specific cutoff may still perform worse than a common cutoff.

Precision/Recall/Enrichment Curves over Thresholds

```
plt <- PRTPlot(df_cm_over_175k,  
              ".fitted", "GT20DWN", 1,  
              plotvars = c("enrichment", "precision", "recall", "specificity", "false_positive_r",  
                           thresholdrange = c(0, threshold_over_175k * 1.25),  
              title = "Enrichment vs. recall with threshold for downpayment model\nVALUE over 175k")  
  
plt +  
  geom_vline(xintercept = threshold_over_175k,  
            color="maroon",  
            linetype = 2)
```

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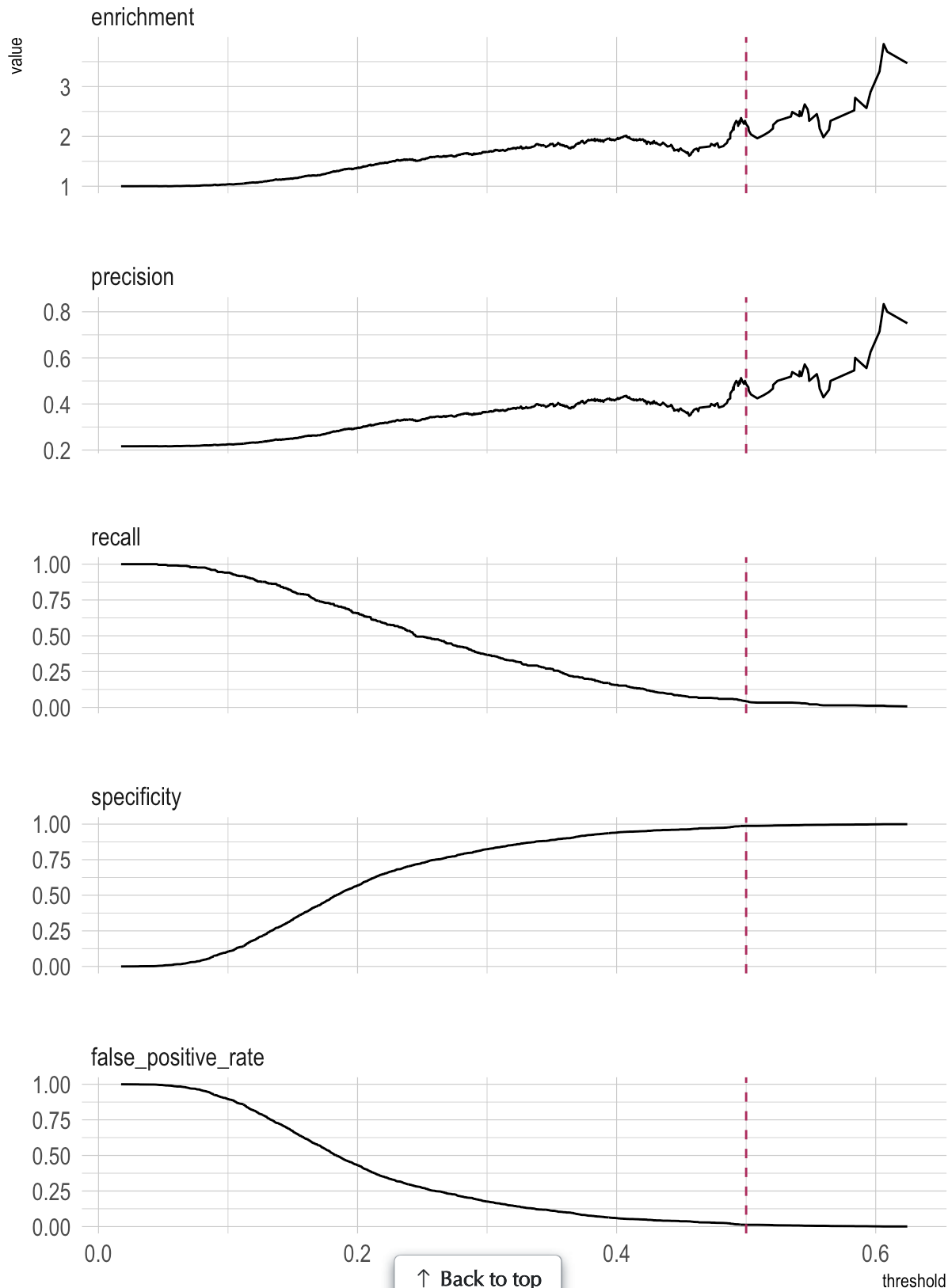
Enrichment vs. recall with threshold for downpayment model VALUE over 175k



```
plt <- PRTPlot(df_cm_under_175k,  
              ".fitted", "GT20DWN", 1,  
              plotvars = c("enrichment", "precision", "recall", "specificity", "false_positive_r  
              thresholdrange = c(0, threshold_under_175k * 1.25),  
              title = "Enrichment vs. recall with threshold for downpayment model\nVALUE under 1  
  
plt +  
  geom_vline(xintercept = threshold_under_175k,  
            color="maroon",  
            linetype = 2)
```

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Enrichment vs. recall with threshold for downpayment model VALUE under 175k



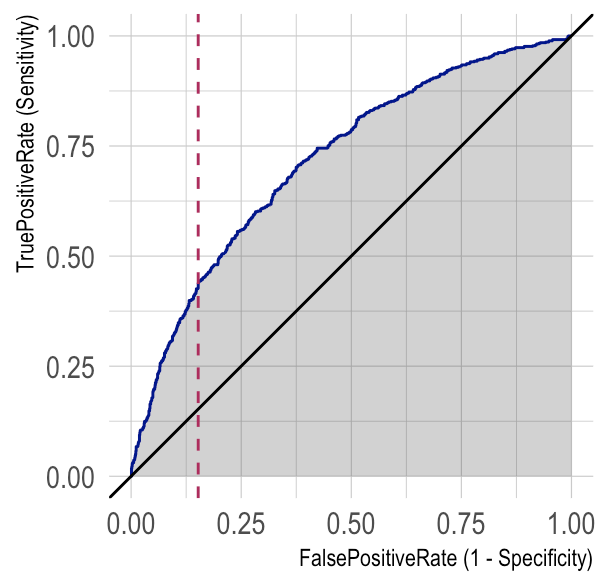
ROC

```
roc <- ROCPlot(df_cm_over_175k,
               xvar = '.fitted',
               truthVar = 'GT20DWN',
               truthTarget = 1,
               title = 'Classifier performance\nVALUE over 175k')

# ROC with vertical line
roc +
  geom_vline(xintercept = 1 - specificity_over_175k,
             color="maroon", linetype = 2)
```

Classifier performance VALUE over 175k GT20DWN==1 ~ .fitted

AUC = 0.72



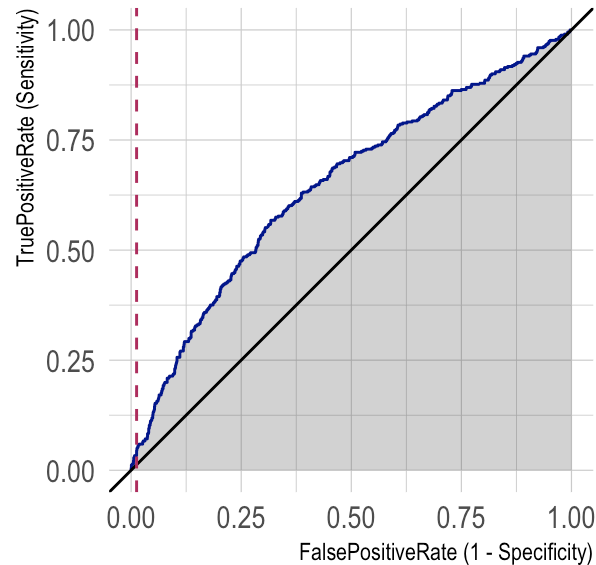
```
roc <- ROCPlot(df_cm_under_175k,
               xvar = '.fitted',
               truthVar = 'GT20DWN',
               truthTarget = 1,
               title = 'Classifier performance\nVALUE under 175k')

# ROC with vertical line
roc +
  geom_vline(xintercept = 1 - specificity_under_175k,
             color="maroon", linetype = 2)
```

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Classifier performance VALUE under 175k GT20DWN==1 ~ .fitted

AUC = 0.65



AUC

```
roc_obj_over_175k <- roc(df_cm_over_175k$GT20DWN, df_cm_over_175k$.fitted)
auc(roc_obj_over_175k)
```

Area under the curve: 0.7177

```
roc_obj_under_175k <- roc(df_cm_under_175k$GT20DWN, df_cm_under_175k$.fitted)
auc(roc_obj_under_175k)
```

Area under the curve: 0.6464

Part 2 Task — Write a blog post (Quarto Document)

Write a blog post about **Part 1 of Homework 2 - Beer Markets**, and add it to your online blog.

Requirements (minimum):

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1. Create a post using a **Quarto Document** (`.qmd`) and ensure it appears on your Quarto website blog.

2. Include:

- a short introduction (what the dataset is and what you'll explore)
- **descriptive statistics** (at least 3 summary numbers)
- **counting** (e.g., top flavors)
- **filtering** (at least one meaningful filter)
- **grouped summaries** (at least one `group_by()` + `summarise()`)
- **regression models** in Homework 1

3. Optional but encouraged:

- at least one ggplot visualization

Deliverables:

- The blog post (in your website repo) renders successfully
- The post is accessible from your website's blog listing page
- In your Brightspace submission (Part 1 QMD), include the **URL** to your published blog post



End of Homework 2

powered with github, quarto, and rstudio
byeong-hak choe, 2026

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